

Post-Quantum Cryptography

for the Novice, the Enjoyer, the Deployer and the Academic

Lena Heimberger

BalCCon 2026, Novi Sad

Cryptography is already a part of your life



messages

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messages



payments

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payments



your car

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your ID

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every connection

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every connection

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It's a skillset, not a separate function.

Mosca's inequality: How long until my secret is revealed?

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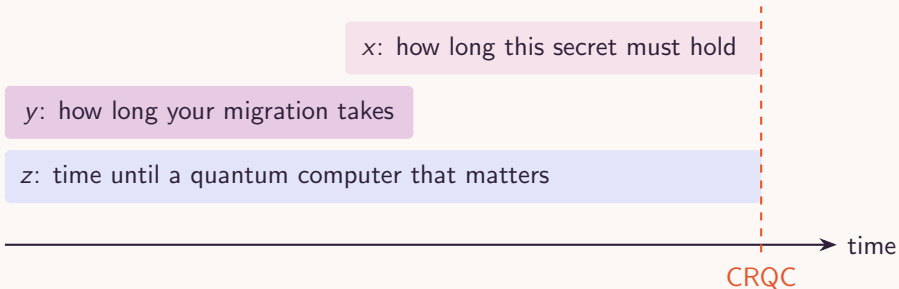
x : how long this secret must hold

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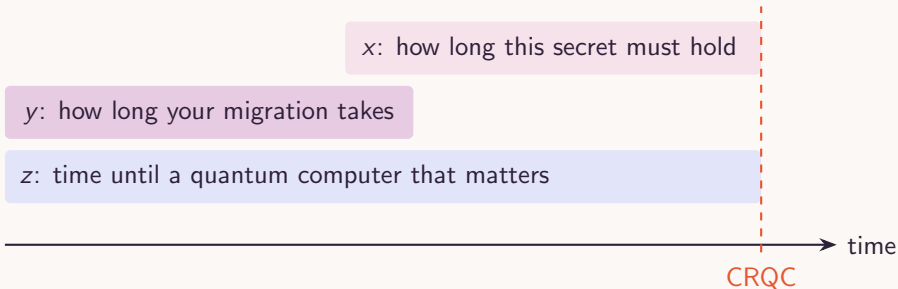
x : how long this secret must hold

y : how long your migration takes

Mosca's inequality: How long until my secret is revealed?



Mosca's inequality: How long until my secret is revealed?



If $x + y > z$, you are already too late.

M. Mosca. *Cybersecurity in an Era with Quantum Computers: Will We Be Ready?* IEEE Security & Privacy 16(5), 2018.

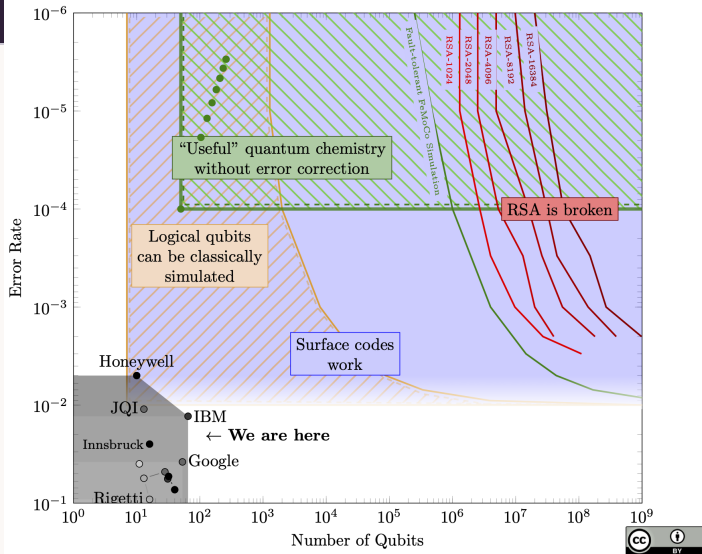


Figure 1: https://sam-jaques.appspot.com/quantum_landscape

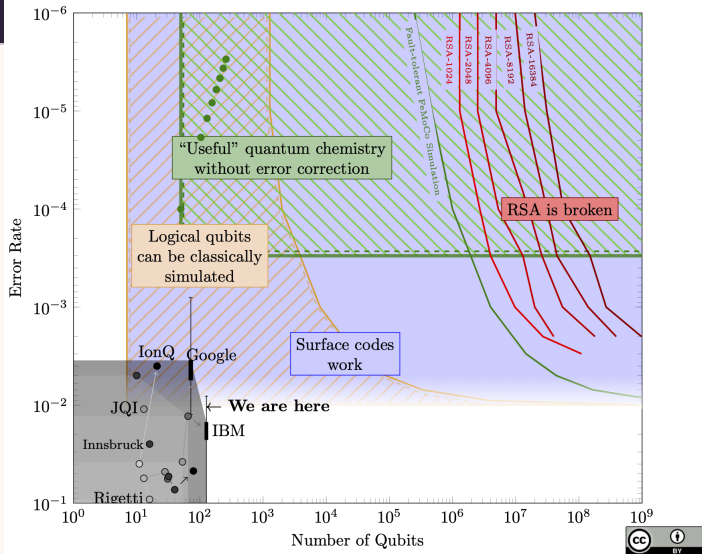


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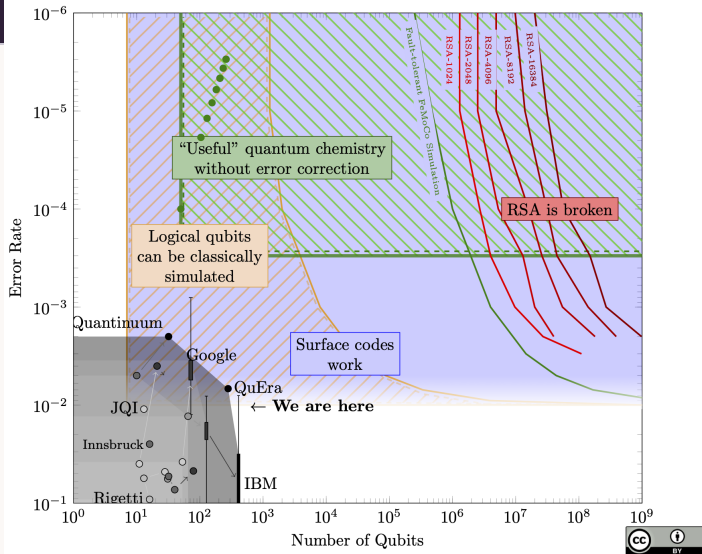


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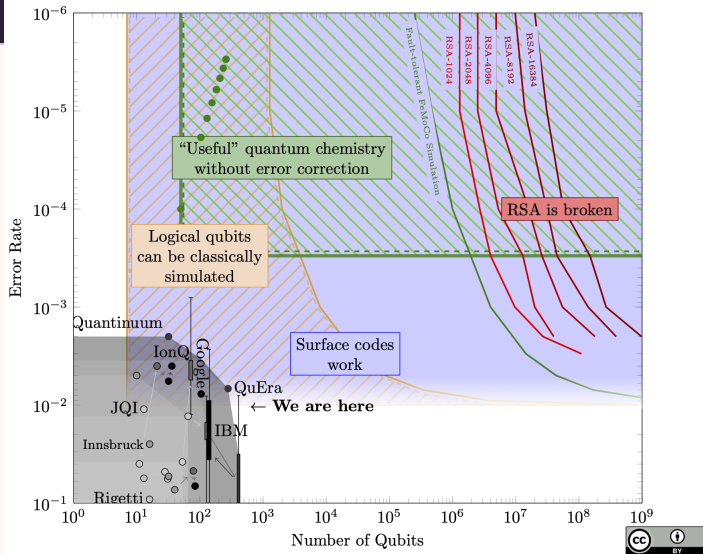


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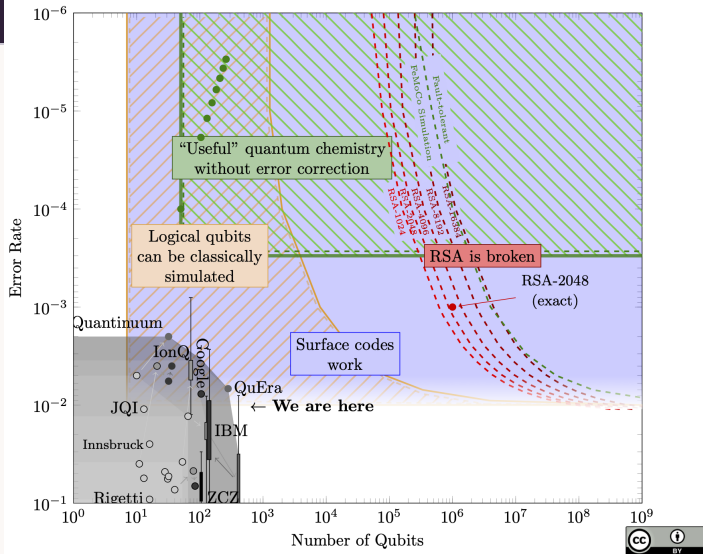


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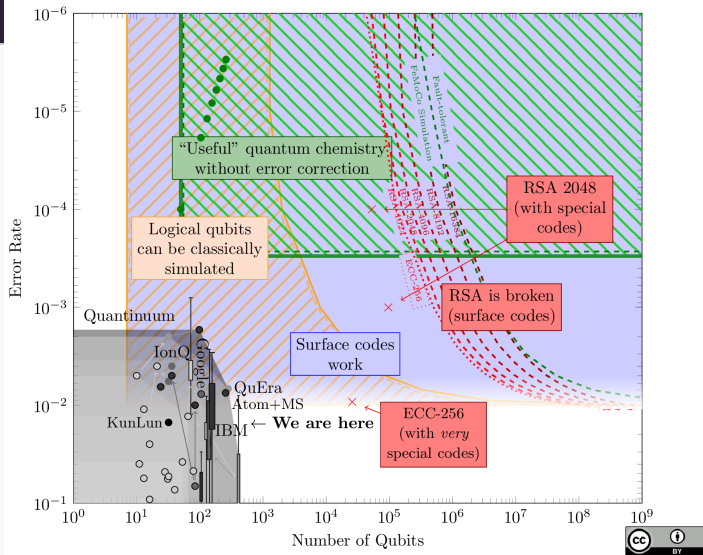
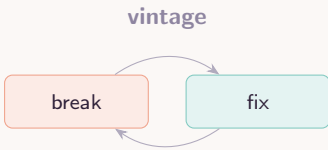
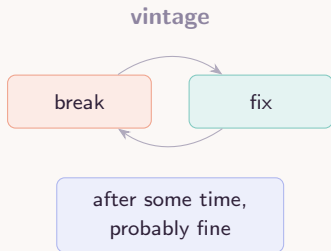


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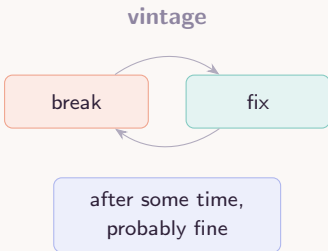
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show breaking the scheme is as hard as breaking underlying assumption

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show breaking the scheme is as hard as breaking underlying assumption
- Reason about model, tightness of reduction, which problem.

Security Levels

NIST level	at least as hard as
1	recovering an AES-128 key
2	finding a SHA-256 collision
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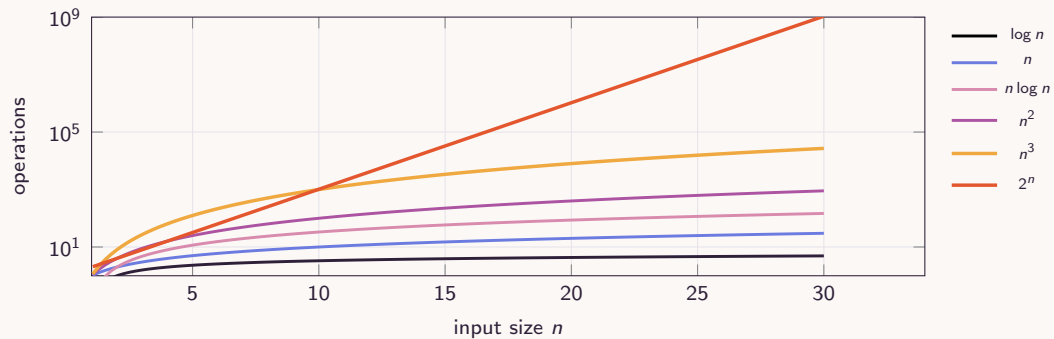
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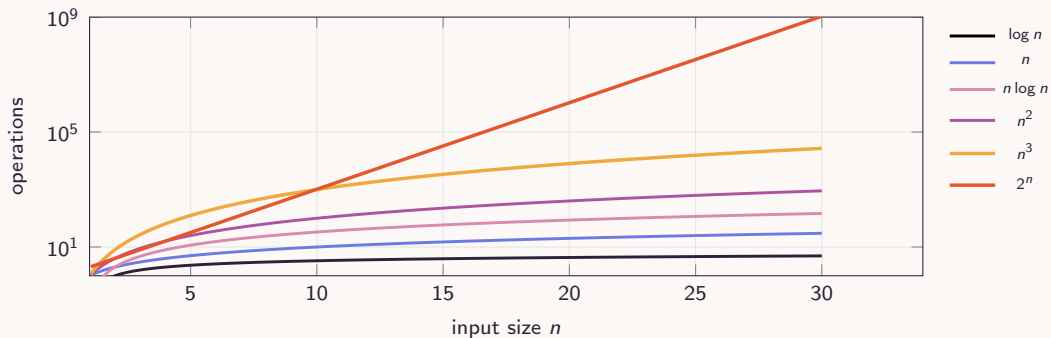
How to factor 2048 bit RSA integers with less than a million noisy qubits, Craig Gidney,

<https://arxiv.org/pdf/2505.15917>

Just double the parameters

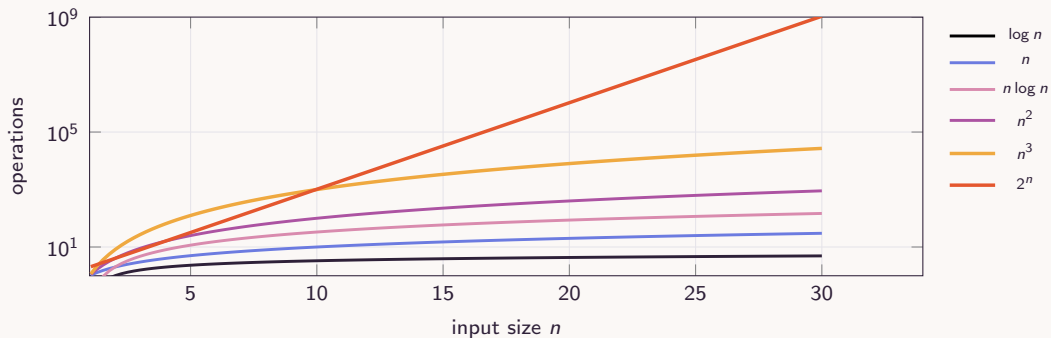


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Also doubles the work you need to do!

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Terabytes of RSA keys

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I also did applied crypto last two summers



[Privacy](#) [Terms](#)

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Talk touches a lot of points. If you ever zone out, just zone back in later.

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Signatures

Forgery with a CRQC.

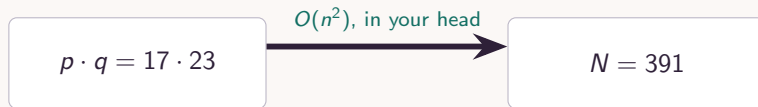
deadline: the day you stop accepting it



the Novice

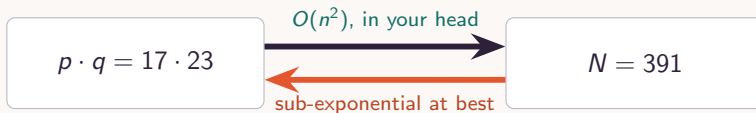
“Why does a quantum computer break cryptography?”

A hard problem becomes easy with a friend's secret



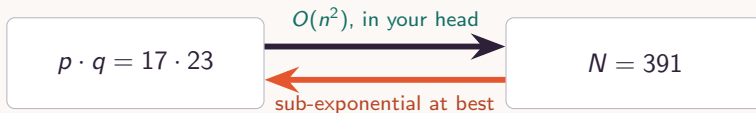
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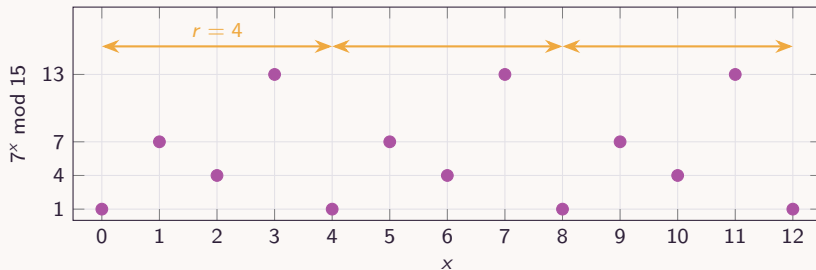
- Polynomial complexity for multiplication.
- general number field sieve runs in $\exp(O((\log N)^{1/3}(\log \log N)^{2/3}))$.
- NP-hard: easy to verify, hard to search.

That bounds the *worst* instance, and cryptography needs the *average* instance to be hard.

Shor reduces factoring to frequency finding

factor N

P. Shor, 1994/1997.

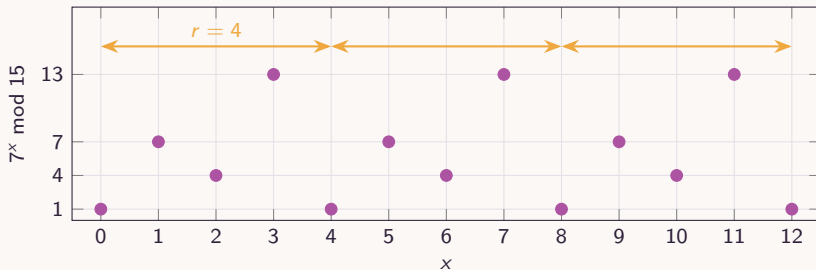


RSA security depends on the attacker not knowing the order of the group.

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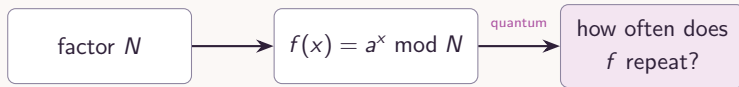


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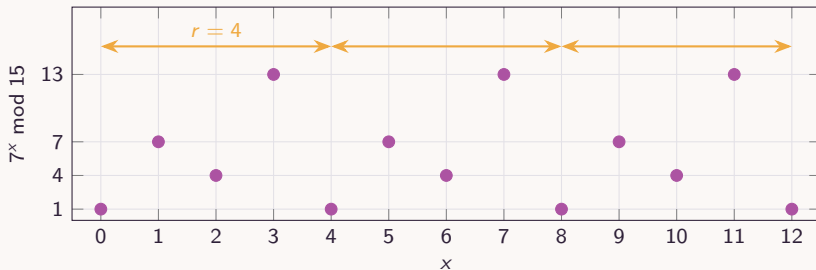


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Blinding operations

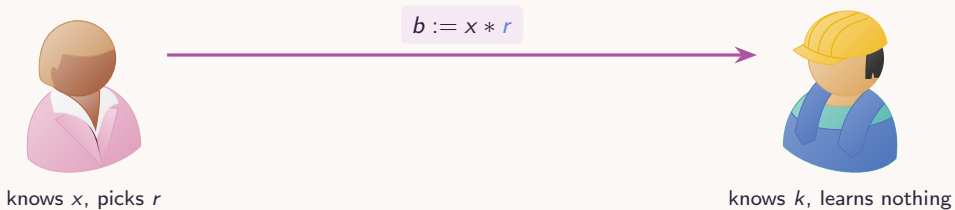


knows x , picks r

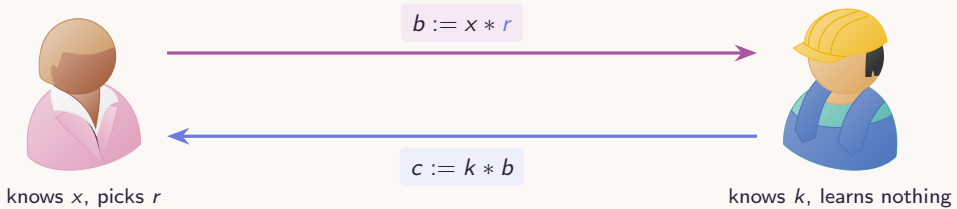


knows k , learns nothing

Blinding operations



Blinding operations



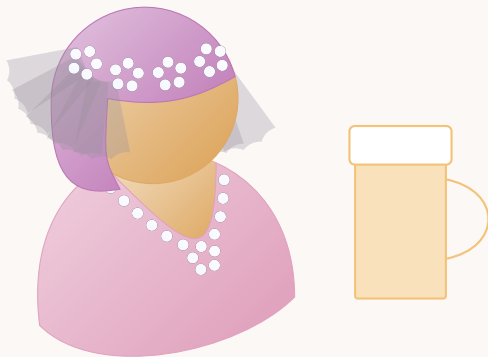
Hidden shifts in commutative settings: Kuperberg



- Same commutative structure, secret moved from *period* to *offset*.
- Then quantum computers have **subexponential** advantage.

G. Kuperberg. A subexponential-time quantum algorithm for the dihedral hidden subgroup problem.
SIAM J. Comput. 35(1), 2005.

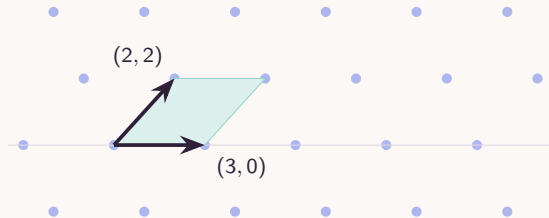
Remove the frequency.



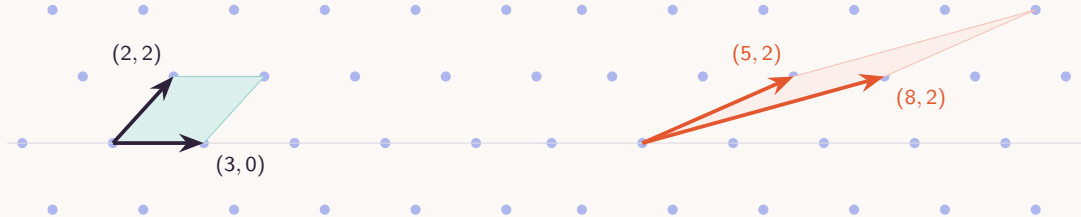
the Enjoyer

“What Alice did last night.”

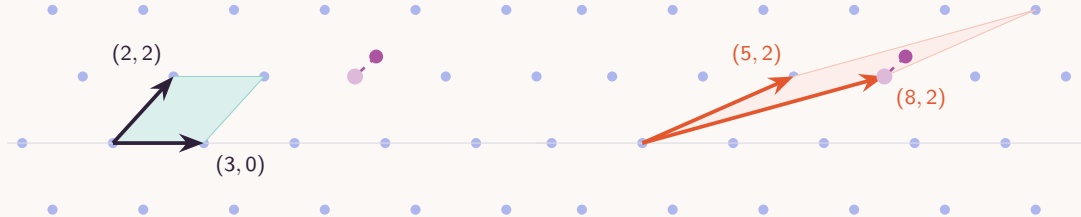
The same lattice, described twice



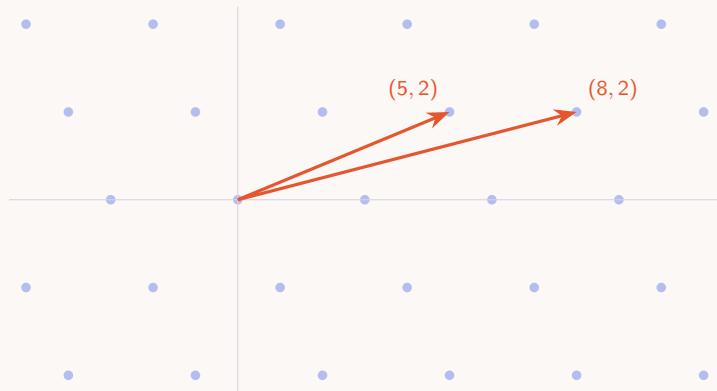
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All about that base



repeat

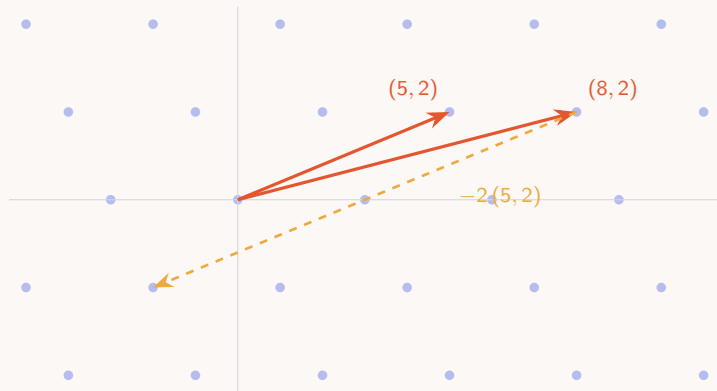
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$$b_2 \leftarrow b_2 - \mu b_1$$

swap if $\|b_2\| < \|b_1\|$

until no swap

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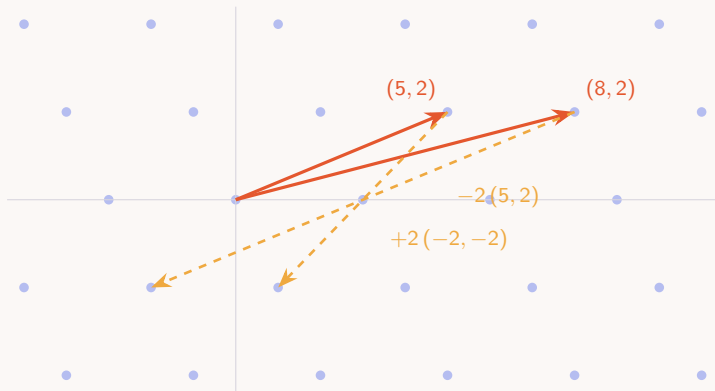
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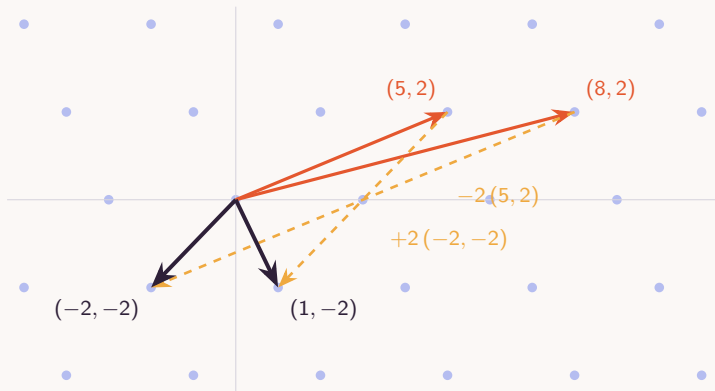
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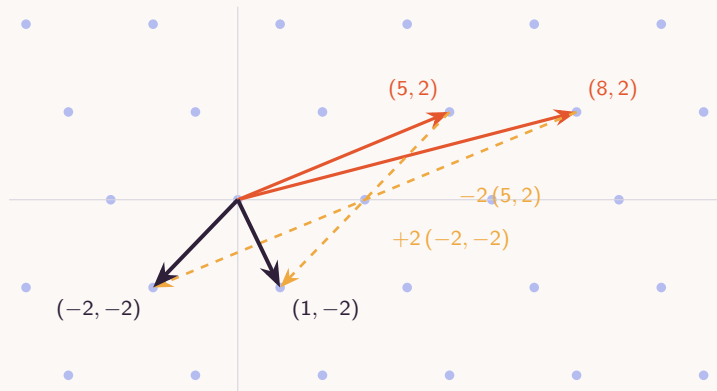
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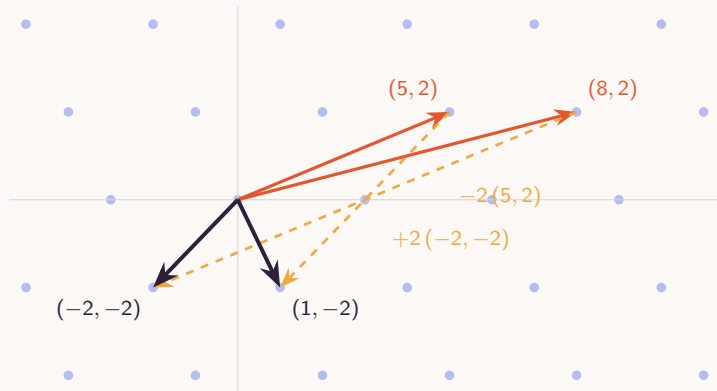
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- LLL is the approximate version.

Lattices: linear algebra with some noise

A diagram illustrating the equation $As = b + e$. The matrix A is represented by a large light blue square. The vector s is a tall, narrow light purple rectangle. The vector e is a tall, narrow light orange rectangle. The vector b is a tall, narrow light teal rectangle. The equation is shown as A followed by s , then a plus sign, then e , then an equals sign, then b . Below each box is descriptive text: A is "public, uniform, big coefficients"; s is "secret, short"; e is "noise, short"; and b is "published".

$$A s + e = b$$

public, uniform,
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Lattices: linear algebra with some noise

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rings and modules to shrink the key, rounding for determinism.

A well-studied problem

- system of univariate linear equations over integers with some noise

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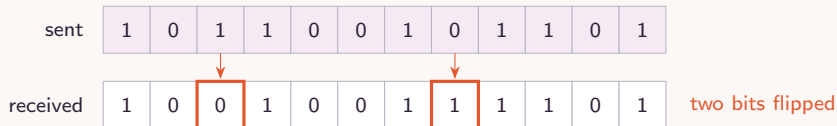
Rise from the ashes of cryptanalysis.

Codes: the same equations, counted differently

sent	1	0	1	1	0	0	1	0	1	1	0	1
received	1	0	0	1	0	0	1	1	1	1	0	1

two bits flipped

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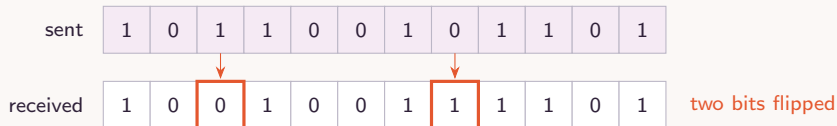
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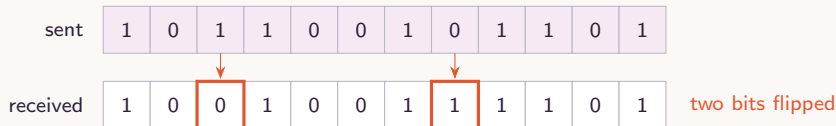
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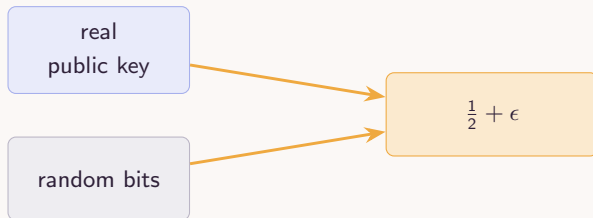
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- ~~conservative choice~~. Distinguisher attack.

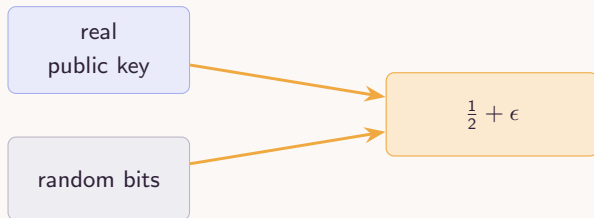
What a distinguisher actually means



Quasipolynomial Cryptanalysis of the McEliece Cryptosystem (or: PIR Meets McEliece)

Ghoshal, Ishai, Jain, Sun, ia.cr/2026/1630

What a distinguisher actually means

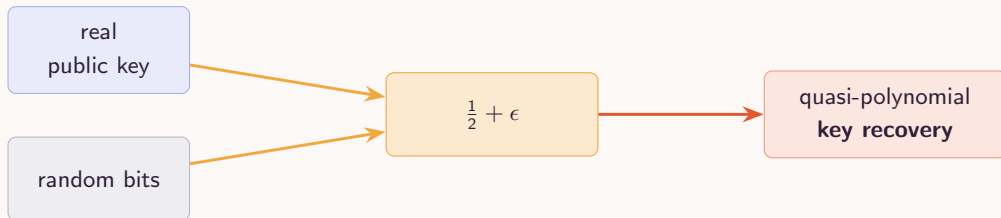


- Why care if someone can tell your public key from noise?

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Ghoshal, Ishai, Jain, Sun, ia.cr/2026/1630

What a distinguisher actually means

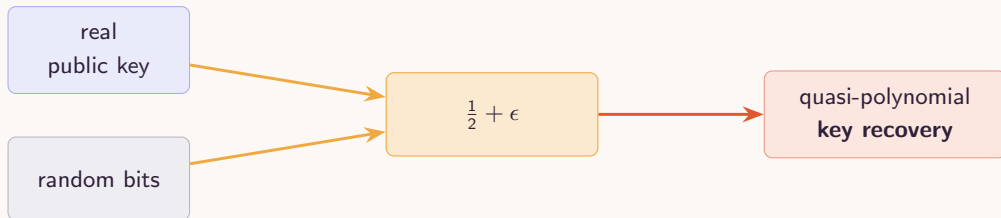


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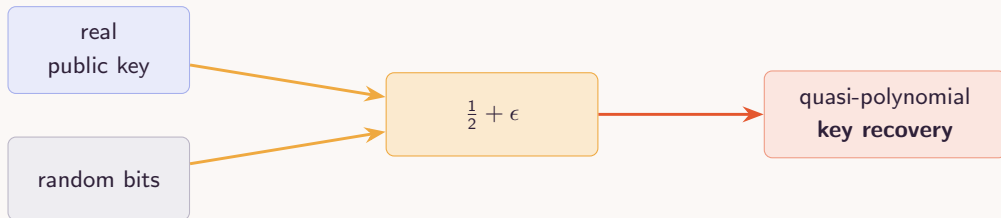


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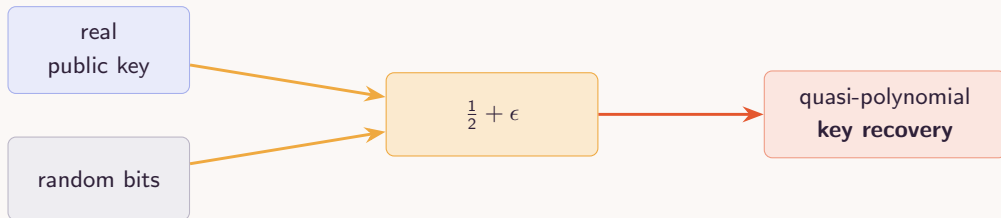


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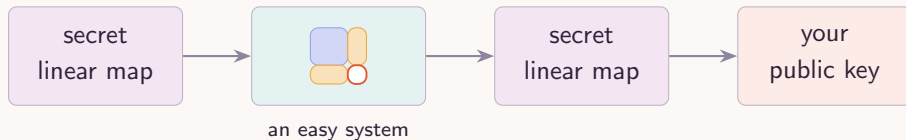


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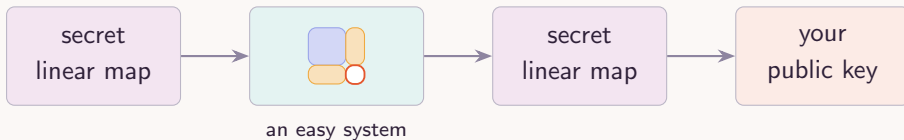
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Multivariate: hide an easy system inside a hard-looking one

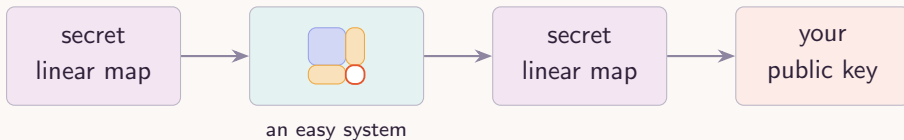


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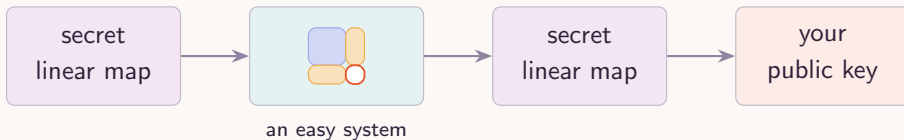
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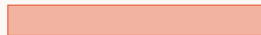
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- 96-byte signatures, public keys in the tens of kilobytes

The easy system: oil and vinegar

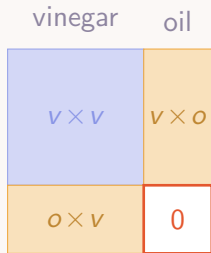
vinegar	oil
$v \times v$	$v \times o$
$o \times v$	0

no oil times oil: fix
the vinegar and the
rest is *linear*

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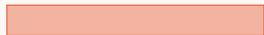


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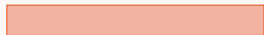
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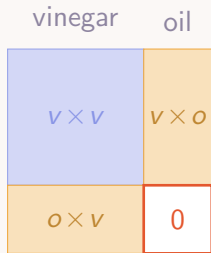
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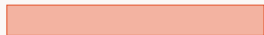
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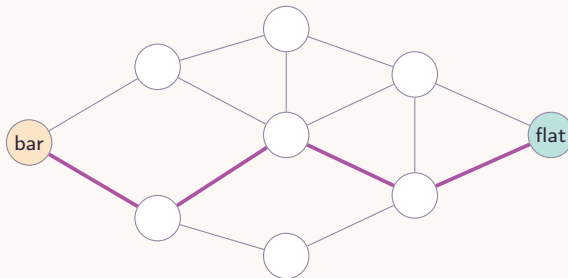
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- Signing: guess the vinegar, solve the rest linearly. Forging without the secret maps: find hidden oil subspace.
- **Balanced** oil and vinegar died in 1998 (Kipnis-Shamir). **Unbalanced**, 2 – 3× more vinegar, still standing
- Shrinking the public key with more structure is risky

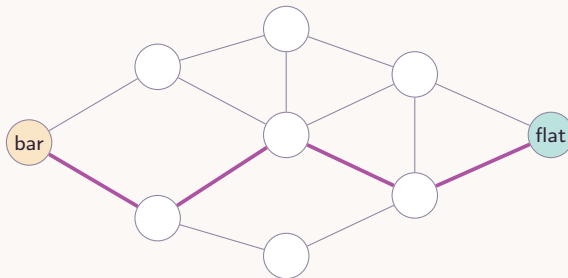
Breaking Rainbow Takes a Weekend on a Laptop, Ward Beullens,

Alice walks home alone at night



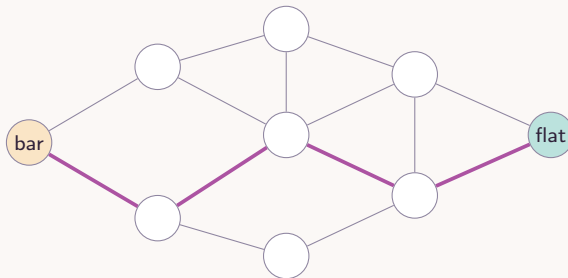
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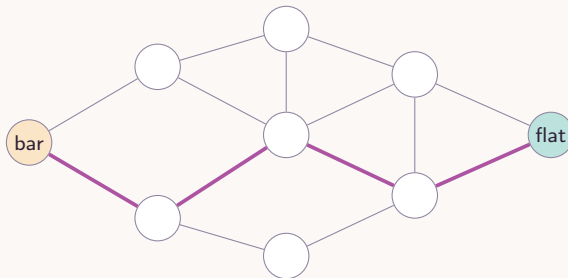
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Isogenies: small, elegant, and a lot of work

- SQIsign: 148-byte signatures

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Public key and signature together fit in a tweet.

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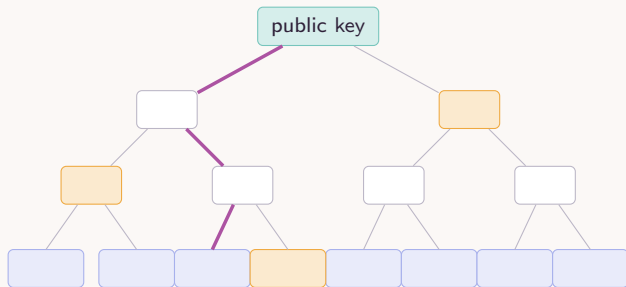
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Public key and signature together fit in a tweet.
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- CSIDH is vulnerable to Kuperberg
- SIDH, the previous favourite, was broken in an afternoon in 2022.

You young people with your fancy math!
We know how to build cryptography.

Symmetric: One way to signatures



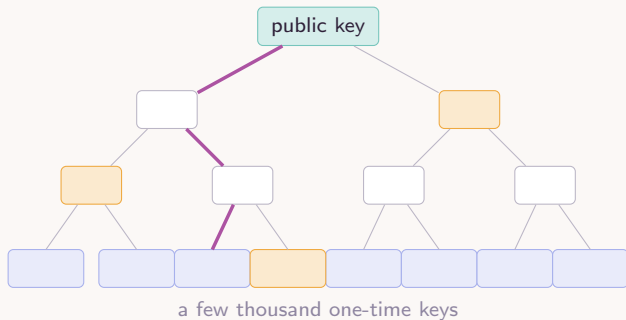
a few thousand one-time keys

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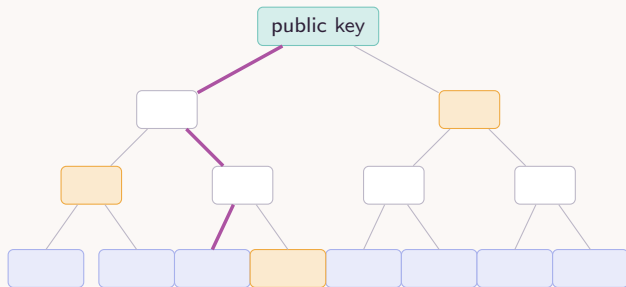
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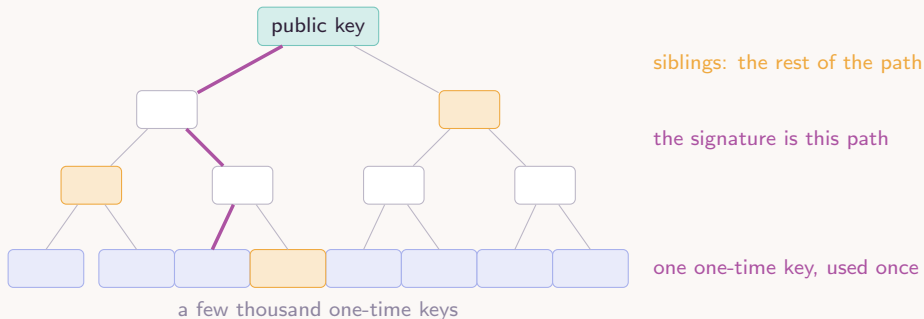
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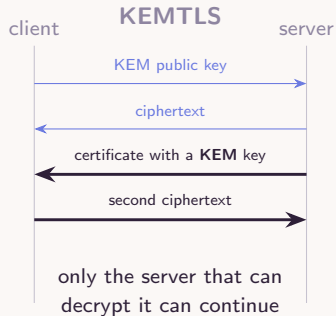
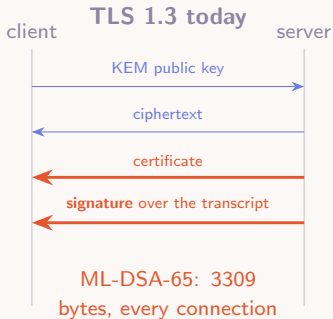
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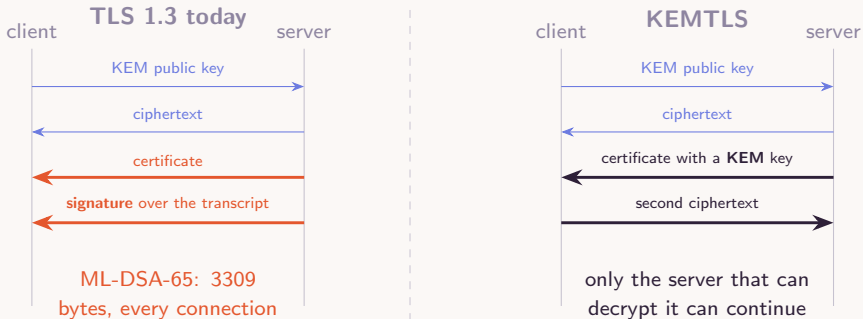
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- Security reduces to underlying hash function. We need one anyways, somewhere

But signatures are expensive!

KEMTLS: authenticate via decryption

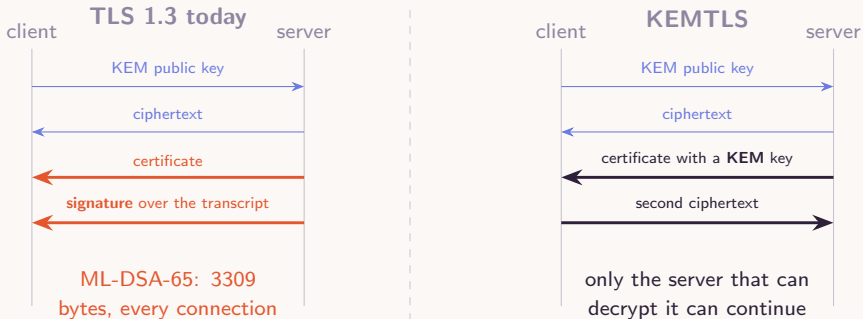


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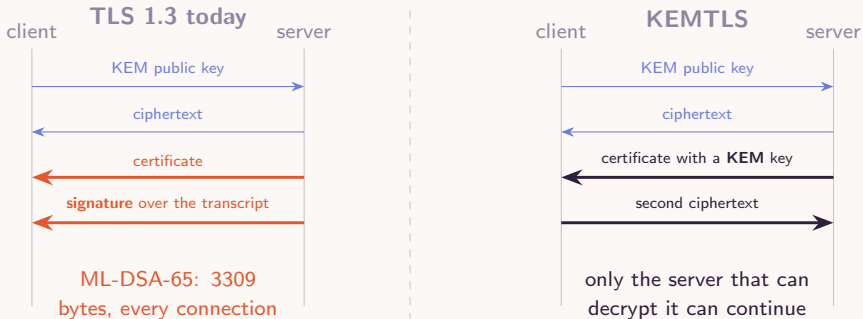
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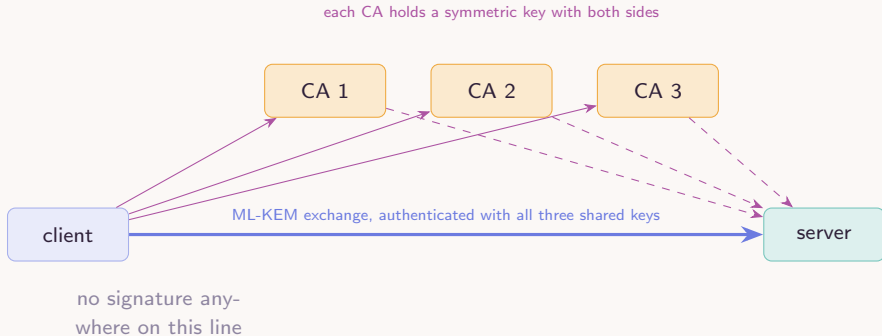
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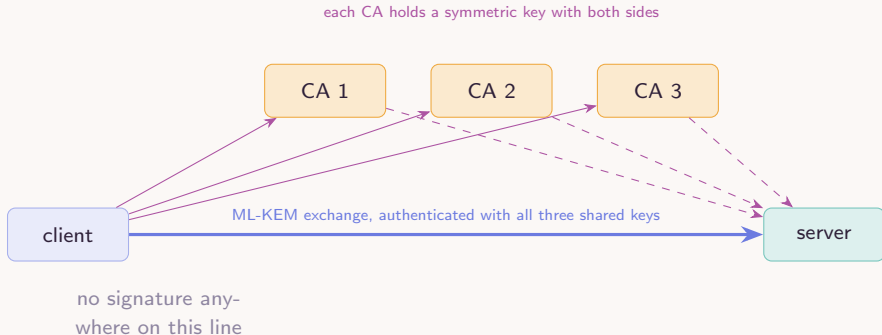


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- Price: authentication is **implicit** at first, and the server's first data waits one extra flight.
- The CA still signs your certificate. KEMTLS removes the one signature made live; Merkle tree certificates remove the four made in advance.

The escape hatch nobody wants: symmetric all the way down

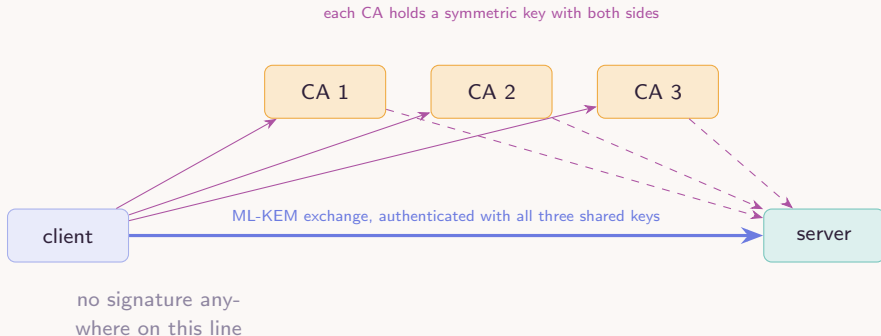


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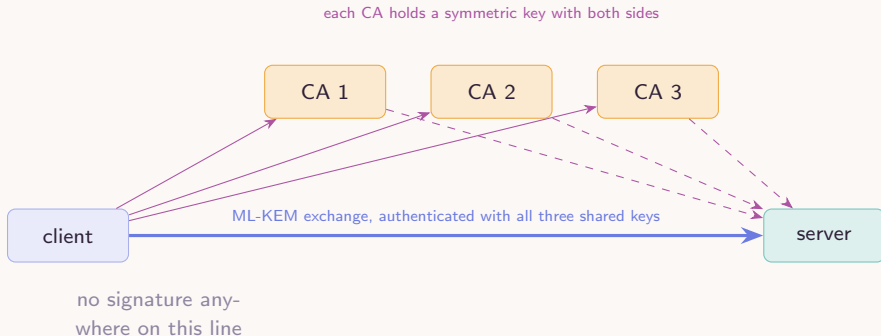
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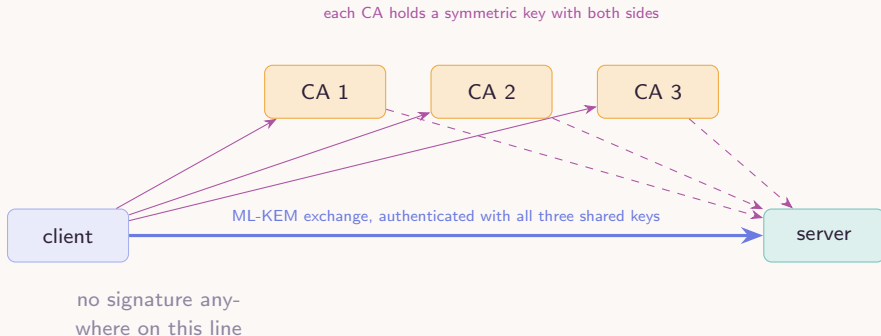
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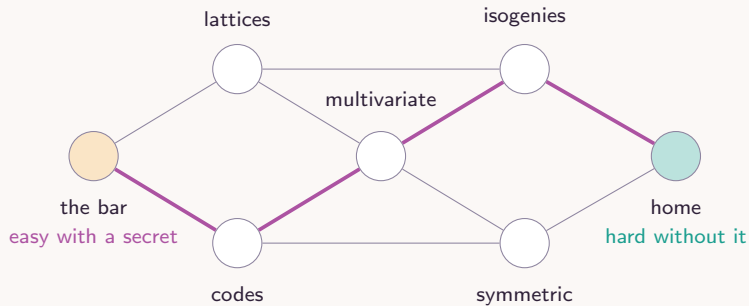
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Adam Langley, <https://www.imperialviolet.org/2024/04/07/letskerberos.html>

Her walk, in hindsight



Where do we go?



the Deployer

“The ethical stage: committing to an algorithm”

The whole menu

standard	what it is	status
ML-KEM (FIPS 203)	key exchange, lattices	shipping, everywhere
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- Everything else is either research prototype or in standardization process

- The internet has decided. Browsers, OpenSSL, Go, Apple, the big CDNs: post-quantum key exchange is on by default and the share only goes up.

Nobody asked you

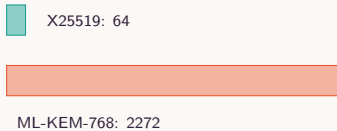
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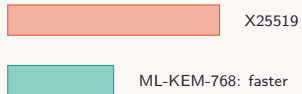
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- the question is *when* rather than *whether*.
- Betting against engineering problems usually means losing.

“But it is bigger”

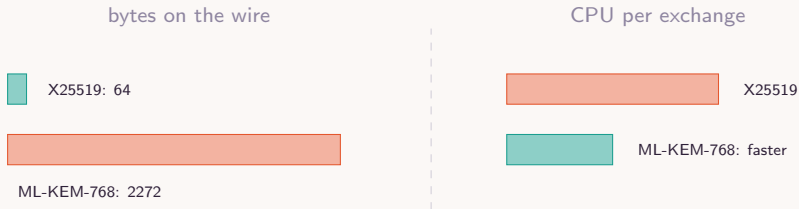
bytes on the wire



CPU per exchange

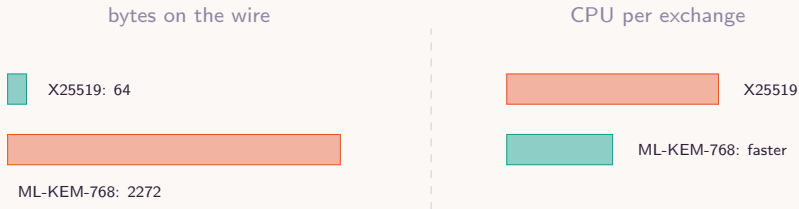


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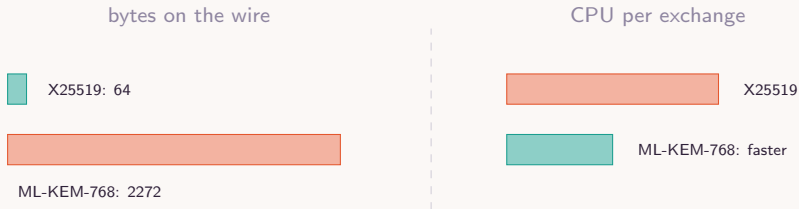
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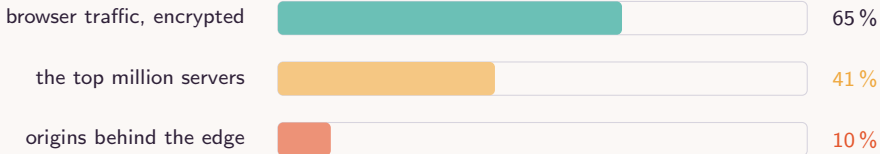
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- if you ship an Electron app tho...

The state of the post-quantum internet

done



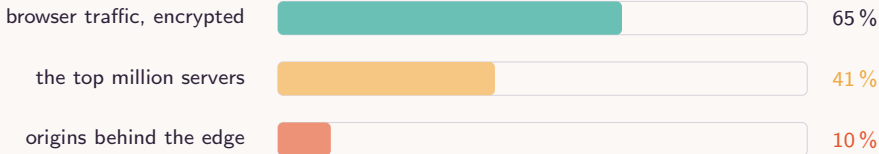
not started

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FN-DSA still draft · hybrid certificate format still unsettled · Merkle tree certificates in testing

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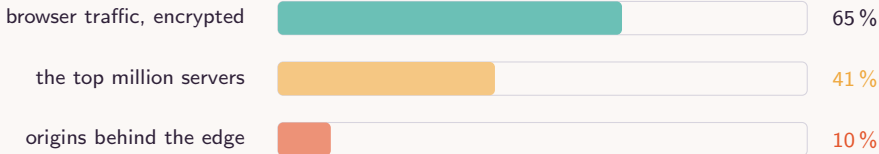
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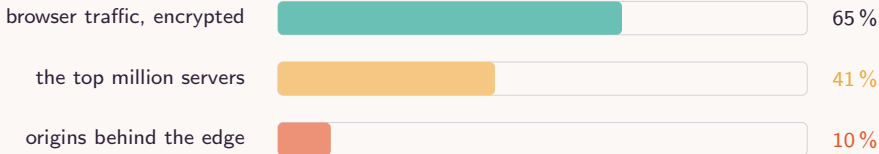
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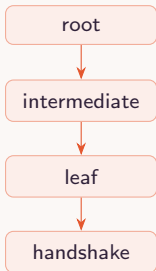
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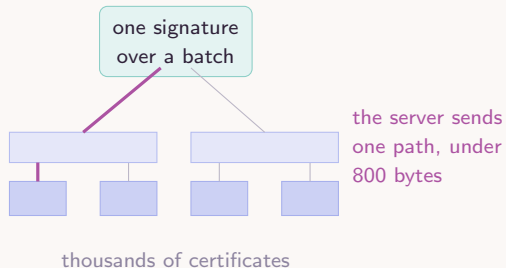
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- Cloudflare and Tranco scans, mid-2026.

If it can be broken, it can be fixed.

Merkle Tree Certificates

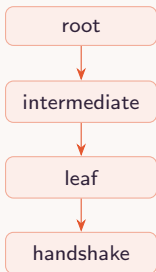


five signatures, sent every time

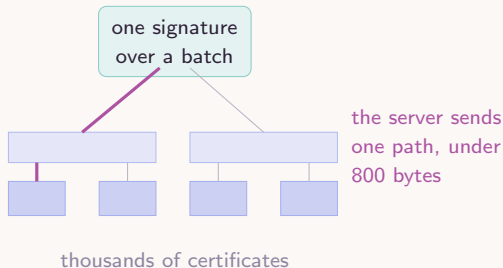


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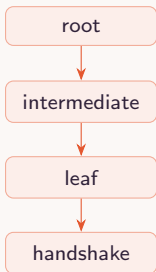


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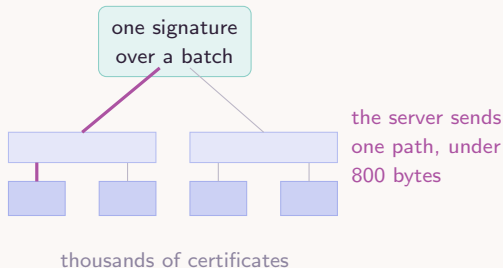


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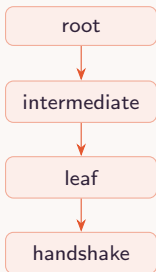


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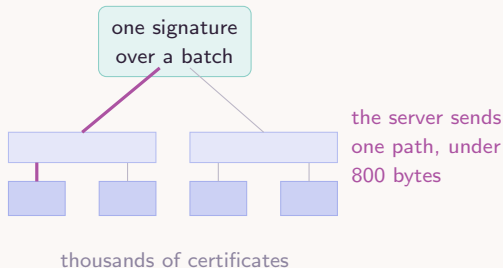


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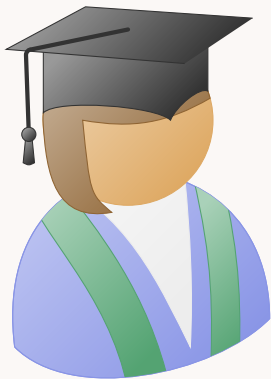
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- PQ+auth could be faster than classical

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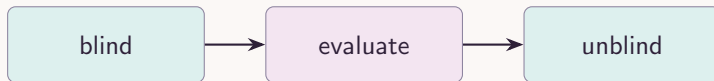
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- A lot of arguments: Hybrids, key sizes, which curve to keep
- Transition is as soon as 2029 (Google, Cloudflare)



the Academic

“What is to be done?”

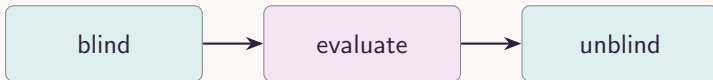
Privacy gets more difficult



$H(x)^r$, then $(\cdot)^k$, then take the r -th root

- Exact, leak-free blinding: Kuperberg.

Privacy gets more difficult



$H(x)^r$, then $(\cdot)^k$, then take the r -th root

- Exact, leak-free blinding: Kuperberg.
- Need to slap on a lot of zero-knowledge proofs and counting arguments.

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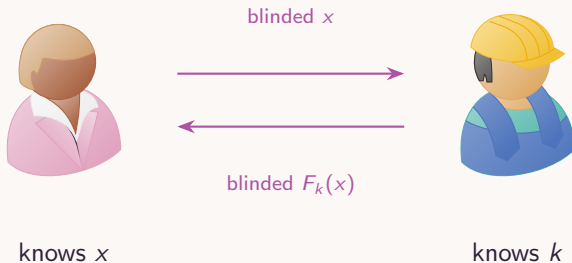
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- **HAWK fell** to a classical attack.
It used structure the other lattices do not have.
- **A quantum algorithm claimed the dihedral coset problem in polynomial time.**
It ended up strengthening lattices: the conjecture is that whole families of quantum lattice attacks may never gain more than a polynomial factor.
- **Classic McEliece got a distinguisher**, which became a quasi-polynomial key recovery. Nobody has run it, much like RSA-1024.

Building to break



- She learns $F_k(x)$. He learns nothing. Neither learns the other's secret.
- "Is my password in this breach?" without saying the password. Private contact discovery. Anonymous tokens.
- As an instrument: the smallest protocol that shows whether a primitive is **flexible**.

What now?



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I am afraid we will lose privacy on the way, because the tools that protect it are the ones that do not survive the move.

And still: the mathematics is beautiful, we get to open systems nobody has opened in thirty years, and almost none of it is decided yet.

Stop being sad.

Break something. On a laptop. Over a weekend.

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Thank you.

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